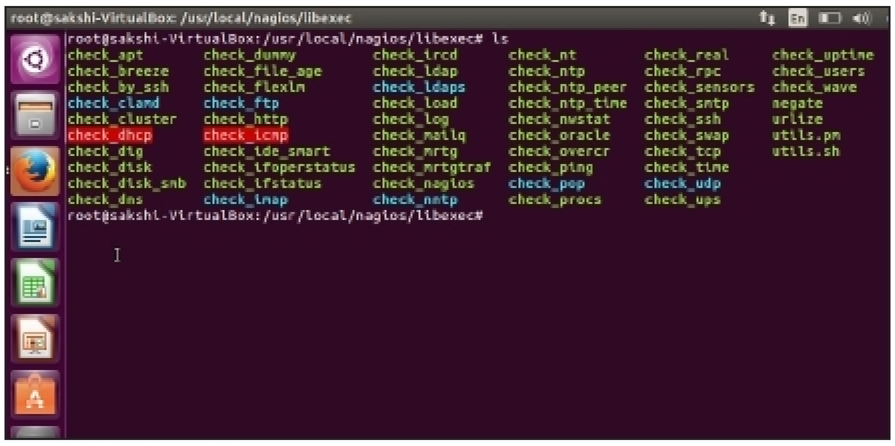

Figure 3: Installed plugins

Restart Nagios Service and reload the Nagios Web page. Now the errors in the host and services page will be resolved as shown in Figure 2. The plugins will now be located in */usr/local/nagios/libexec/* (see Figure 3).

Writing a custom plugin

Nagios plugins can be written in any language. Based on the exit code, Nagios decides the status of the service. Here is an example of a custom plugin to check the status of the Apache service running locally. The program written to check the status of this service should be placed in */usr/local/nagios/libexec*. As this program can be written in any language, the code for it is not shown here. Let us consider it to be a shell script, *check_apache.sh*. To run this check as a command, it should be defined in */usr/local/nagios/etc/objects/commands.cfg* as:

```
# 'check_apache' command definition
define command{
      command_name check_apache
      command_line $USER1$/check_apache.sh
}
```

$USER1$ variable is a local Nagios variable pointing to */usr/local/nagios/libexec*. One can associate this command to a service, and then to a host by defining it in */usr/local/nagios/etc/objects/localhost.cfg*.

```
# Define a service to check apache on local machine.
define service{
      use                     local-service
      host_name               localhost
      service_description     apache
      check_command           check_apache
      }
```

Now check whether the configuration is free from all errors and restart the Nagios service by running the following commands:

```
$sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
$sudo systemctl reload-or-restart nagios.service
```

After the execution, the status will be reflected in the Web UI.

The benefits of Nagios

Nagios is one of the best open source monitoring tools available in the market. It is modular and has a straightforward approach, which makes it easy to use. One can add functionalities to meet specific needs. Nagios can work with other tools like *Check_mk* and Vigilo NMS to provide the user with additional functionalities. This makes Nagios very flexible. It can be easily scaled to a large number of hosts and to a large network. Nagios uses compilers and binaries as well as scripts, the utility of which is facilitated by an ingenious abstraction layer. END

References

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